

SECTION 3 SIGNAL DESCRIPTION

This section contains brief descriptions of the input and output signals in their functional groups, as shown in Figure 3-1. Each signal is explained in a brief paragraph with reference to other sections that contain more detail about the signal and the related operations.

NOTE

In this section and in the remainder of the manual, *assert* and *negate* are used to specify forcing a signal to a particular state. In particular, *assertion* and *assert* refer to a signal that is active or true; *negation* and *negate* indicate a signal that is inactive or false. These terms are used independently of the voltage level (high or low) that they represent.

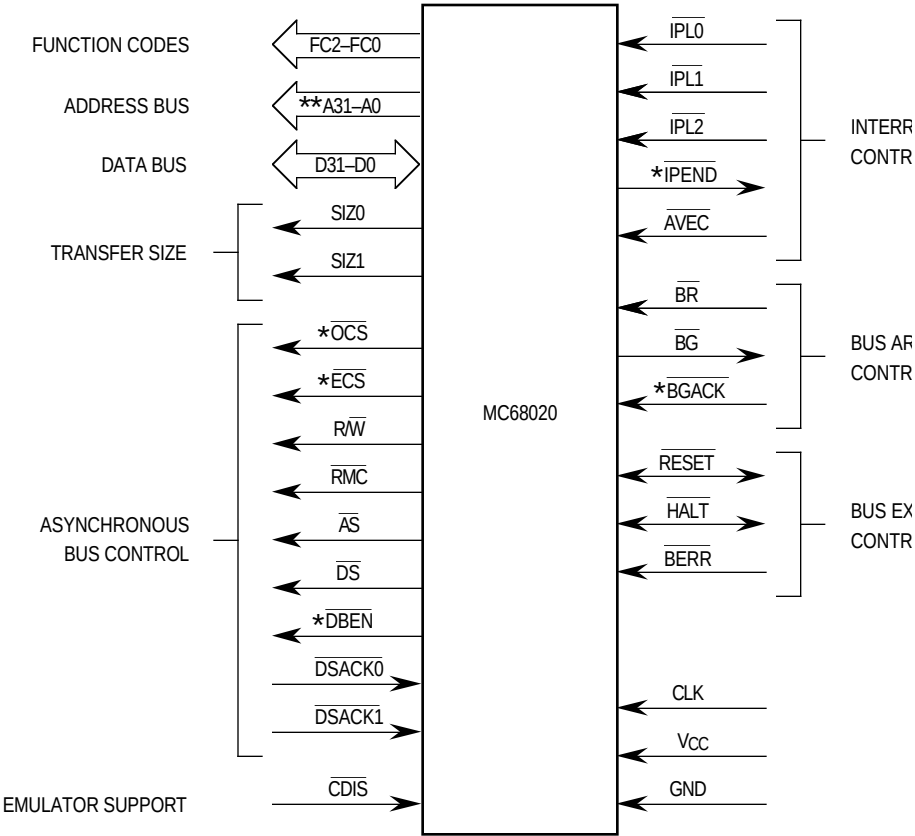


Figure 3-1. Functional Signal Groups

3.1 SIGNAL INDEX

The input and output signals for the MC68020/EC020 are listed in Table 3-1. Both the names and mnemonics are shown along with brief signal descriptions. Signals that are implemented in the MC68020, but not in the MC68EC020, have an asterisk (*) preceding the signal name in Table 3-1. Also, note that the address bus is 32 bits wide for the MC68020 and 24 bits wide for the MC68EC020. For more detail on each signal, refer to the paragraph in this section named for the signal and the reference in that paragraph to a description of the related operations.

Timing specifications for the signals listed in Table 3-1 can be found in **Section 10 Electrical Characteristics**.

3.2 FUNCTION CODE SIGNALS (FC2–FC0)

These three-state outputs identify the address space of the current bus cycle. Table 2-1 shows the relationships of the function code signals to the privilege levels and the address spaces. Refer to **Section 2 Processing States** for more information.

3.3 ADDRESS BUS (A31–A0, MC68020)(A23–A0, MC68EC020)

These three-state outputs provide the address for the current bus cycle, except in the CPU address space. Refer to **Section 2 Processing States** for more information on the CPU address space. A31 is the most significant address signal for the MC68020; A23 is the most significant address signal for the MC68EC020. The upper eight bits (A31–A24) are used internally by the MC68EC020 to access the internal instruction cache address tag. Refer to **Section 5 Bus Operation** for information on the address bus and its relationship to bus operation.

3.4 DATA BUS (D31–D0)

These three-state bidirectional signals provide the general-purpose data path between the MC68020/EC020 and all other devices. The data bus can transfer 8, 16, 24, or 32 bits of data per bus cycle. D31 is the most significant bit of the data bus. Refer to **Section 5 Bus Operation** for more information on the data bus and its relationship to bus operation.

3.5 TRANSFER SIZE SIGNALS (SIZ1, SIZ0)

These three-state outputs indicate the number of bytes remaining to be transferred for the current bus cycle. Signals A1, A0, DSACK1, DSACK0, SIZ1, and SIZ0 define the number of bits transferred on the data bus. Refer to **Section 5 Bus Operation** for more information on SIZ1 and SIZ0 and their use in dynamic bus sizing.

Table 3-1. Signal Index

Signal Name	Mnemonic	Function
Function Codes	FC2–FC0	3-bit function code used to identify the address space of each bus cycle.
Address Bus MC68020 MC68EC020	A31–A0 A23–A0	32-bit address bus 24-bit address bus
Data Bus	D31–D0	32-bit data bus used to transfer 8, 16, 24, or 32 bits of data per bus cycle.
Size	SIZ1, SIZ0	Indicates the number of bytes remaining to be transferred for this cycle. These signals, together with A1 and A0, define the active sections of the data bus.
*External Cycle Start	ECS	Provides an indication that a bus cycle is beginning.
*Operand Cycle Start	OCS	Identical operation to that of ECS except that OCS is asserted only during the first bus cycle of an operand transfer.
Read/Write	R/W	Defines the bus transfer as a processor read or write.
Read-Modify-Write Cycle	RMC	Provides an indicator that the current bus cycle is part of an indivisible read-modify-write operation.
Address Strobe	AS	Indicates that a valid address is on the bus.
Data Strobe	DS	Indicates that valid data is to be placed on the data bus by an external device or has been placed on the data bus by the MC68020/EC020.
*Data Buffer Enable	DBEN	Provides an enable signal for external data buffers.
Data Transfer and Size Acknowledge	DSACK1, DSACK0	Bus response signals that indicate the requested data transfer operation has completed. In addition, these two lines indicate the size of the external bus port on a cycle-by-cycle basis and are used for asynchronous transfers.
Interrupt Priority Level	IPL2–IPL0	Provides an encoded interrupt level to the processor.
*Interrupt Pending	IPEND	Indicates that an interrupt is pending.
Autovector	AVEC	Requests an autovector during an interrupt acknowledge cycle.
Bus Request	BR	Indicates that an external device requires bus mastership.
Bus Grant	BG	Indicates that an external device may assume bus mastership.
*Bus Grant Acknowledge	BGACK	Indicates that an external device has assumed bus mastership.
Reset	RESET	System reset.
Halt	HALT	Indicates that the processor should suspend bus activity or that the processor has halted due to a double bus fault.
Bus Error	BERR	Indicates that an erroneous bus operation is being attempted.
Cache Disable	CDIS	Statically disables the on-chip cache to assist emulator support.
Clock	CLK	Clock input to the processor.
Power Supply	V _{CC}	Power supply.
Ground	GND	Ground connection.

*This signal is implemented in the MC68020 and not implemented in the MC68EC020.

3.6 ASYNCHRONOUS BUS CONTROL SIGNALS

The following signals control synchronous bus transfer operations for the MC68020/EC020. Note that OCS, ECS, and DBEN are implemented in MC68020 and not implemented in the MC68EC020.

Operand Cycle Start (OCS, MC68020 only)

This output signal indicates the beginning of the first external bus cycle for an instruction prefetch or a data operand transfer. OCS is not asserted for subsequent cycles that are performed due to dynamic bus sizing or operand misalignment. Refer to **Section 5 Bus Operation** for information about the relationship of OCS to bus operation.

OCS is not implemented in the MC68EC020.

External Cycle Start (ECS, MC68020 only)

This output signal indicates the beginning of a bus cycle of any type. Refer to **Section 5 Bus Operation** for information about the relationship of ECS to bus operation.

ECS is not implemented in the MC68EC020.

Read/Write (R/W)

This three-state output signal defines the type of bus cycle. A high level indicates a read cycle; a low level indicates a write cycle. Refer to **Section 5 Bus Operation** for information about the relationship of R/W to bus operation.

Read-Modify-Write Cycle (RMC)

This three-state output signal identifies the current bus cycle as part of an indivisible read-modify-write operation; it remains asserted during all bus cycles of the read-modify-write operation. Refer to **Section 5 Bus Operation** for information about the relationship of RMC to bus operation.

Address Strobe (AS)

This three-state output signal indicates that a valid address is on the address bus. The FC2–FC0, SIZ1, SIZ0, and R/W signals are also valid when AS is asserted. Refer to **Section 5 Bus Operation** for information about the relationship of AS to bus operation.

Data Strobe (DS)

During a read cycle, this three-state output signal indicates that an external device should place valid data on the data bus. During a write cycle, DS indicates that the MC68020/EC020 has placed valid data on the bus. During two-clock synchronous write cycles, the MC68020/EC020 does not assert DS. Refer to **Section 5 Bus Operation** for more information about the relationship of DS to bus operation.

Data Buffer Enable (DBEN, MC68020 only)

This output signal is an enable signal for external data buffers. This signal may not be required in all systems. Refer to **Section 5 Bus Operation** for more information about the relationship of DBEN to bus operation.

DBEN is not implemented in the MC68EC020.

Data Transfer and Size Acknowledge (DSACK1, DSACK0)

These input signals indicate the completion of a requested data transfer operation. In addition, they indicate the size of the external bus port at the completion of each cycle. These signals apply only to asynchronous bus cycles. Refer to **Section 5 Bus Operation** for more information on these signals and their relationship to dynamic bus sizing.

3.7 INTERRUPT CONTROL SIGNALS

The following signals are the interrupt control signals for the MC68020/EC020. Note that IPEND is implemented in the MC68020 and not implemented in the MC68EC020.

Interrupt Priority Level Signals (IPL2–IPL0)

These input signals provide an indication of an interrupt condition and the encoding of the interrupt level from a peripheral or external prioritizing circuitry. IPL2 is the most significant bit of the level number. For example, since the IPL2–IPL0 signals are active low, IPL2–IPL0 equal to \$5 corresponds to an interrupt request at interrupt level 2. Refer to **Section 6 Exception Processing** for information on MC68020/EC020 interrupts.

Interrupt Pending (IPEND, MC68020 only)

This output signal indicates that an interrupt request exceeding the current interrupt priority mask in the SR has been recognized internally. This output is for use by external devices (coprocessors and other bus masters, for example) to predict processor operation on the following instruction boundaries. Refer to **Section 6 Exception Processing** for interrupt information. Also, refer to **Section 5 Bus Operation** for bus information related to interrupts.

IPEND is not implemented in the MC68EC020.

Autovector (AVEC)

This input signal indicates that the MC68020/EC020 should generate an automatic vector during an interrupt acknowledge cycle. Refer to **Section 5 Bus Operation** for more information about automatic vectors.

3.8 BUS ARBITRATION CONTROL SIGNALS

The following signals are the bus arbitration control signals used to determine which device in a system is the bus master. Note that BGACK is implemented in the MC68020 and not implemented in the MC68EC020.

Bus Request (BR)

This input signal indicates that an external device needs to become the bus master. BR is typically a “wire-ORed” input (but does not need to be constructed from open-collector devices). Refer to **Section 5 Bus Operation** for more information on MC68020 bus arbitration. Refer to **Section 5 Bus Operation** and **Appendix A Interfacing an MC68EC020 to a DMA Device That Supports a Three-Wire Bus Arbitration Protocol** for more information on MC68EC020 bus arbitration.

Bus Grant (BG)

This output signal indicates that the MC68020/EC020 will release ownership of the bus when the current processor bus cycle completes. Refer to **Section 5 Bus Operation** for more information on MC68020 bus arbitration. Refer to **Section 5 Bus Operation** and **Appendix A Interfacing an MC68EC020 to a DMA Device That Supports a Three-Wire Bus Arbitration Protocol** for more information on MC68EC020 bus arbitration.

Bus Grant Acknowledge (BGACK, MC68020 only)

This input signal indicates that an external device has become the bus master. Refer to **Section 5 Bus Operation** for more information on MC68020 bus arbitration. Refer to **Section 5 Bus Operation** and **Appendix A Interfacing an MC68EC020 to a DMA Device That Supports a Three-Wire Bus Arbitration Protocol** for more information on MC68EC020 bus arbitration.

BGACK is not implemented in the MC68EC020.

3.9 BUS EXCEPTION CONTROL SIGNALS

The following signals are the bus exception control signals for the MC68020/EC020.

Reset (RESET)

This bidirectional open-drain signal is used to initiate a system reset. An external reset signal resets the MC68020/EC020 as well as all external devices. A reset signal from the processor (asserted as part of the RESET instruction) resets external devices only; the internal state of the processor is not altered. Refer to **Section 5 Bus Operation** for a description of reset bus operation and **Section 6 Exception Processing** for information about the reset exception.

Halt (HALT)

The assertion of this bidirectional open-drain signal indicates that the processor should suspend bus activity or, when used with BERR, that the processor should retry the current cycle. Refer to **Section 5 Bus Operation** for a description of the effects of HALT on bus operations. When the processor has stopped executing instructions due to a double bus fault condition, the HALT line is asserted by the processor to indicate to external devices that the processor has stopped.

Bus Error (BERR)

This input signal indicates that an invalid bus operation is being attempted or, when used with HALT, that the processor should retry the current cycle. Refer to **Section 5 Bus Operation** for a description of the effects of BERR on bus operations.

3.10 EMULATOR SUPPORT SIGNAL

The following signal supports emulation by providing a means for an emulator to disable the on-chip cache by supplying internal status information to an emulator. Refer to **Section 7 Coprocessor Interface Description** for more detailed information on emulation support.

Cache Disable (CDIS)

This input signal statically disables the on-chip cache to assist emulator support. Refer to **Section 4 On-Chip Cache Memory** for information about the cache; refer to **Section 9 Applications Information** for a description of the use of this signal by an emulator. CDIS does not flush the instruction cache; entries remain unaltered and become available again when CDIS is negated.

3.11 CLOCK (CLK)

The CLK signal is the clock input to the MC68020/EC020. This TTL-compatible signal should not be gated off at any time while power is applied to the processor. Refer to **Section 9 Applications Information** for suggestions on clock generation. Refer to **Section 10 Electrical Characteristics** for electrical characteristics.

3.12 POWER SUPPLY CONNECTIONS

The MC68020/EC020 requires connection to a V_{CC} power supply, positive with respect to ground. The V_{CC} connections are grouped to supply adequate current for the various sections of the processor. The ground connections are similarly grouped. **Section 11 Ordering Information and Mechanical Data** describes the groupings of V_{CC} and ground connections, and **Section 9 Applications Information** describes a typical power supply interface.

3.13 SIGNAL SUMMARY

Table 3-2 provides a summary of the characteristics of the signals discussed in this section. Signal names preceded by an asterisk (*) are implemented in the MC68020 and not implemented in the MC68EC020.

Table 3-2. Signal Summary

Signal Function	Signal Name	Input/Output	Active State	Three-State
Function Codes	FC2–FC0	Output	High	Yes
Address Bus MC68020 MC68EC020	A31–A0 A23–A0	Output	High	Yes
Data Bus	D31–D0	Input/Output	High	Yes
Transfer Size	SIZ1, SIZ0	Output	High	Yes
*Operand Cycle Start	OCS	Output	Low	No
*External Cycle Start	ECS	Output	Low	No
Read/Write	R/W	Output	High/Low	Yes
Read-Modify-Write Cycle	RMC	Output	Low	Yes
Address Strobe	AS	Output	Low	Yes
Data Strobe	DS	Output	Low	Yes
*Data Buffer Enable	DBEN	Output	Low	Yes
Data Transfer and Size Acknowledge	DSACK1, DSACK0	Input	Low	—
Interrupt Priority Level	IPL2–IPL0	Input	Low	—
*Interrupt Pending	IPEND	Output	Low	No
Autovector	AVEC	Input	Low	—
Bus Request	BR	Input	Low	—
Bus Grant	BG	Output	Low	No
*Bus Grant Acknowledge	BGACK	Input	Low	—
Reset	RESET	Input/Output	Low	No **
Halt	HALT	Input/Output	Low	No **
Bus Error	BERR	Input	Low	—
Cache Disable	CDIS	Input	Low	—
Clock	CLK	Input	—	—
Power Supply	V _{CC}	Input	—	—
Ground	GND	Input	—	—

*This signal is implemented in the MC68020 and not implemented in the MC68EC020.

** Open-drain